Why Moran and Half Edge Diverge

Hypotheses and Test Plan

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1 The puzzle

Visually, across the metro data, dissimilarity falls, Moran's I rises, and Half Edge falls slightly (or if we zoom in op top 10 populated metro areas, if falls similarly to dissimilarity). The purpose of this memo is to lay out possible explanations for that divergence and the tests that would tell us whether each explanation is right.

A useful distinction throughout the memo is the difference between showing that a mechanism *can* produce divergence and showing that it *did* produce the historical divergence:

1. Changing only the proposed driver moves the metrics in the predicted directions.
2. The proposed driver actually changed between 1980 and 2020.

2 What counts as divergence?

2.1 A few ideas for measuring trend divergence

First, there's the eyeball test: looking at yearly lineplots we can easily see the trends. This may be enough for many tests below. But we can also have a proper definition of divergence between the metrics.

Moran and Half Edge are not naturally on the same scale. A tentative solution is to divide changes in each metric by its range. For metro i in year t, define

$$D_{it}^{\text{trend}} = \frac{M_{it} - M_{i,1980}}{R_M} - \frac{H_{it} - H_{i,1980}}{R_H}, \tag{1}$$

where M is Moran, H is Half Edge, and R_M and R_H are their ranges. A positive value means that Moran has risen more, or fallen less, than Half Edge since 1980. Or maybe use since t-1.

There are two reasonable choices for the ranges:

- **Theoretical ranges.** If the Moran implementation is treated as having range $[-1, 1]$, then $R_M = 2$. However, we know that it can have a larger range, so this is not ideal. (Half Edge has range $[0, 1]$). Under that choice, Moran levels can be mapped to $[0, 1]$. We should verify that $[-1, 1]$ is the appropriate bound though.

- **Or use the observed range.** Use the pooled maximum minus minimum across the metro-year data.

2.2 Percentile disagreement

This one does not depend on the metrics having comparable numerical scales and just asks if M and HE rank metros differently in a given year.

Suppose a year contains N metros. Sort their Moran values from lowest to highest and assign each metro a rank r_M. Do the same for Half Edge to get r_H. Since we have different number of CBSAs by year, we will convert each rank to a percentile: $P_M = \frac{r_M - 1}{N-1}$ (or just $\frac{r_M}{N}$), and $P_H = \frac{r_H - 1}{N-1}$, (or just $\frac{r_H}{N}$).

The lowest-ranked metro has percentile 0 and the highest-ranked metro has percentile 1. Define disagreement as $P_{M,it} - P_{H,it}$.

3 Hypothesis 1: unequal tract populations

Moran gives every tract equal importance, while Half Edge gives more influence to tracts containing more people. Real tract populations vary greatly, while what we saw on theoretical graphs gives every node the same population, there's no variation in population.

Test: equal-population counterfactual

For each tract, preserve its Black share and its neighbors, but set its total population to 1,000. If tract i has Black share p_i, the counterfactual counts are

$$X_i^* = 1000p_i, \qquad Y_i^* = 1000(1 - p_i). \tag{2}$$

Recompute Moran, Half Edge, and dissimilarity on every counterfactual graph. Reproduce the trend graphs, and if it's needed after the eyeball test, we calculate both definitions of divergence.

The eyeball test was negative (looking at trends), we don't observe that.

4 Hypothesis 2: binary versus mixed tracts

Toy graphs classify each node as one group or the other, but real tracts contain shares. The metrics may agree on binary graphs but behave differently on real, mixed tracts.

Test: the assortativity trend

Use the assortativity metric. Assortativity provides is a binary Half Edge. The code would calculate yearly metro means for assortativity and plot its trend next to Moran, Half Edge, and dissimilarity.

As a visual test: no, assortativety doesn't get rid of this trend.

5 Hypothesis 3: composition amplitude

Two maps can have the same spatial pattern even if one has much larger differences between tracts. Moran standardizes those differences away. Dissimilarity and Half Edge do not. To test this, we define amplitude as the difference between the mean ρ of the metro and the ρ of a tract.

Test 1: scale amplitude down

For each metro graph, move every tract share toward the metro-wide share:

$$p_i(\alpha) = \bar{p} + \alpha(p_i - \bar{p}). \tag{3}$$

Here $\alpha = 1$ is the observed graph. Its smaller values compress the tract differences. Tract populations and adjacency remain fixed. The code recomputes the metrics over a grid of α values (from 0 to 1). Moran should remain fixed for positive α, dissimilarity should fall, and Half Edge should move toward its neutral value.

It is indeed what we observe.

Test 2: define and measure amplitude in the real data

It is important to decide what "amplitude" means before using it as a historical explanation.

The code would calculate this quantity for every CBSA-year, plot its distribution and mean by year, and maybe plot within-CBSA trajectories. (This appears to be flat, so that's probably not the explanation of trends.)

Test 3: connect amplitude change to divergence

For metros observed in 1980 and 2020, calculate ΔA_i and the change in Moran–Half Edge divergence (defined above, one way or another). Plot the relationship and estimate a regression? If amplitude explains the historical divergence, metros where amplitude fell more should show a larger increase in divergence.

The tentative amplitude measure does not fall over time, so apparently amplitude can explain a difference in metric general "level" but not the historical trend.

6 Hypothesis 4: cluster diffusion

A concentrated Black population may experience diffusion into nearby tracts. Neighboring tracts can then become more similar, which raises Moran (and higher perimeter raises Moran), while the population becomes less unevenly concentrated (more diffused), which lowers dissimilarity and may lower Half Edge.

(Diffusion can reduce composition and amplitude at the same time. How to split them?)

Test 1: diffuse a cluster on the Iowa graph

We start with a concentrated Black cluster on the Iowa tract graph. At each step, transfer a small amount of Black population from a higher-share tract to lower-share neighboring tracts. (Move White population in the opposite direction so that every tract total and the statewide Black and White totals remain fixed?) Recompute metrics after every step and save.

Test 2: repeat the diffusion algorithm on theoretical graphs? If needed

Run the same algorithm on simple theoretical graphs. Regular grids and a small set of graph shapes with different boundary and degree structures. Repeat the experiment from several starting clusters, vary the placement of the starting point.

This tells us whether the Iowa result is general or depends on one particular graph. After all, Iowa is apparently notoriously weird.

Test 3: decide how to measure diffusion in real CBSAs

To test the hypothesis historically, we need a definition of diffusion that can be calculated for every real CBSA. And plot it to see any changes over time.

Test 4: relate real diffusion to metric divergence

Once a diffusion measure is chosen, calculate it for every CBSA-year and estimate a panel model such as

$$D_{it}^{\text{trend}} = \alpha_i + \gamma_t + \beta\,\text{Diffusion}_{it} + \theta A_{it} + \delta \rho_{it} + \eta\,\text{PopulationCV}_{it} + \kappa\,\text{Topology}_{it} + \varepsilon_{it}. \qquad (4)$$

Here α_i is a CBSA fixed effect and γ_t is a year fixed effect (do we need them though?.. we can have them). Standard errors clustered by CBSA. The coefficient of interest is β. The amplitude (A) control is here because diffusion and amplitude may move together.

7 Hypothesis 6: graph topology

Moran and the between-tract part of Half Edge both use adjacency, but they use it differently. Changes in which tracts count as neighbors may therefore move the metrics by different amounts.

Test 1: direct edge rewiring

Hold every tract's population counts fixed and generate alternative edge sets using edge swaps. Degree-preserving. Keep each graph connected. Recompute Moran and Half Edge for many rewired versions, dissimilarity should remain unchanged.

8 Hypothesis 7: observed changes in graph size and structure

Some of the change in the metrics may come from changes in the tract graphs themselves. Metros have different numbers of nodes and edges in 1980 and 2020, and those changes may be related to metric change.

Give each metro the following three variables:

$$\Delta \text{metric}_i = \text{metric}_{i,2020} - \text{metric}_{i,1980}, \tag{5}$$
$$\Delta \#\text{edges}_i = \#\text{edges}_{i,2020} - \#\text{edges}_{i,1980}, \tag{6}$$
$$\Delta \#\text{nodes}_i = \#\text{nodes}_{i,2020} - \#\text{nodes}_{i,1980}. \tag{7}$$

Maybe calculate the change in leaves as well, and run a regression or something. A simple version is

$$\Delta \text{metric}_i = \alpha + \beta_E \Delta \#\text{edges}_i + \beta_N \Delta \#\text{nodes}_i + \beta_L \Delta \#\text{leaves}_i + \varepsilon_i. \tag{8}$$

The code would run this separately for Moran, Half Edge, and the standardized Moran–Half Edge divergence. It would produce coefficient plots and simple scatterplots so the result can be assessed visually as well as through the regression.

9 Combined counterfactual analysis

After the individual tests, it would be useful to include one combined counterfactual analysis. For each CBSA, sequentially remove:

1. tract population heterogeneity;
2. the observed amplitude change;
3. within-tract weighting;

4. the observed topology;
5. the change in metro composition ρ.

After each change, recompute Moran, Half Edge, dissimilarity, and divergence measure(s?).

These mechanisms interact, so the result can depend on the order in which they are removed. The code can repeat the exercise in several reasonable orders.

10 Final evidence table

Fill in as the tests are completed:

Hypothesis	Mechanism works?	Driver changed?	Within-CBSA link?	Verdict
Equal populations				
Binary versus mixed tracts				
Amplitude				
Cluster diffusion				
Within-tract inflation				
Topology				
Graph size and structure				
Changing ρ				